

Reflective Activity A — Generative AI Impact in Software Security in Emails

Introduction

Targeted email attacks cost less and persuade better since large language models arrived. Attackers now build believable baits in many languages on a large scale. They copy voices and add personal touches from gathered details. For IT experts, the issue goes beyond tech. It covers rules, user faith, and job duties. I stand firm: view email safety as a mix of social and tech parts. Strengthen tools and staff with rule-based steps, clear filters, and ongoing training (Correa et al., 2023; Deckard, 2023).

Context and governance landscape

Enterprises rely on layered defences: domain authentication (SPF, DKIM, DMARC), secure email gateways, endpoint protection, and user training. Generative AI alters attacker economics, so controls need a governance refresh. Professional duties under recognised codes, and ethics frameworks such as Menlo, require attention to consent, fairness, transparency, and accountability in any automated screening pipeline (Finn and Shilton, 2023). Policy should set risk appetite, define acceptable use of GenAI for outbound messaging, and mandate logging for audit and redress (Mittelstadt, 2019; UNESCO, 2021).

Case comparison

Think of two places. A school takes in more outside mail and prizes open links. A clinic holds high needs for privacy and steady service. Both use checks and blocks but set different limits and alerts. At school, hard blocks hurt teamwork. At clinic, loose rules bring danger. AI-made phishing shifts the balance: changing text seems real, so blocks must blend words, sender trust, action signs, and user flags. In both spots, apparent reasons count workers want quick, simple notes on holds, and checkers need steady records (Deckard, 2023; Floridi, 2019).

Recommended course of action.

I suggest a six-step plan that weighs safety against ease:

- Rules and oversight. Okay a short policy on email and GenAI use stop auto-fill tricks for outside mail without checks; say when staff can use GenAI for answer drafts; demand tags on AI help; set levels for holds or mild alerts. Share a brief 'care duty' note with users.
- Checks and origins. Force DMARC with block on owned names; watch outside links and push them to SPF/DKIM. Mark unsigned mail from risky ends as odd by default.
- Blocks with openness. Pick tools that give part-based reasons to checkers—like odd word choices, name mismatches, strange send habits. Show users a short bar with clear causes, not just a score, to aid smart choices (Mittelstadt, 2019).
- Human checks in reports. Make 'flag odd' one tap in all email apps; open a sort of list auto with full items kept. Send results back to the tool as marked info; share three-month stats on hits, misses, and wrong flags.
- Drills that match risks. Do quick tests with true-to-life, job-fit baits. Track gains over time; skip blame focus. Offer fast, small lessons when users tap bars or flag items.
- Prep for events and fixes. Hold guides for account grabs, pay shift tricks, and fake bills. Log choices, tell hit users, and give a plain path to fight auto-block results.

Legal, social, and professional implications

Large-scale email scans start data guard tasks: need, fit, and short keeps. Hold times should split logs from full texts. Apparent reasons aid fair play—users should know block causes and appeal steps. On the job, tech workers must note build picks, list weak spots, and skip big claims on catch rates (BCS Code of Conduct; Menlo principles). In groups, faith grows when staff spot open steps, quick replies, and firm help in events (UNESCO, 2021).

Risks and mitigations

Main dangers cover tool shifts, wrong flags that break tasks, and gaps in other tongues. Fixes: steady tests on new sets; wide sample groups; two-level alerts that note not stop unsure cases; checked skips for key jobs; and outside checks of learn data (Correa et al., 2023).

Reflective close

In my work, I will firm up rule changes for blocks, add brief reason notes to each hold, and plan three-month talks with non-security groups—HR, money, and ops. This roots the setup in true chat flows we guard for pay (Deckard, 2023; Mittelstadt, 2019).

Reflective close

In my own practice, I will tighten change control around filtering rules, add short rationale strings to every quarantine event, and schedule quarterly reviews with stakeholders outside security—HR, finance, and operations. This keeps the system grounded in the real communication patterns we are paid to protect (Deckard, 2023; Mittelstadt, 2019).

References

BCS – The Chartered Institute for IT (n.d.) Code of Conduct. Available at: [IT Apprenticeships | BCS](#)

Correa, J.R., Cows, J., Tsamados, A., Taddeo, M. and Floridi, L. (2023) 'Global AI governance: a framework for comparing international AI policies', *AI and Ethics*, 3, 155–172. [A Unified Framework of Five Principles for AI in Society · Issue 1.1, Summer 2019](#)

Deckard, M. (2023) 'Ethics in the age of Generative AI: balancing innovation and responsibility', *Communications of the ACM*, 66(5), 24–27. [\(PDF\) Ethical Considerations in the Age of Artificial Intelligence: Balancing Innovation and Social Values](#)

Finn, M. and Shilton, K. (2023) 'Ethics governance development: The case of the Menlo Report', *Social Studies of Science*, 53(3), 315–340.

Floridi, L. (2019) 'Establishing the rules for building trustworthy AI', *Nature Machine Intelligence*, 1(6), 261–262.

Mittelstadt, B.D. (2019) 'Principles alone cannot guarantee ethical AI', *Nature Machine Intelligence*, 1(11), 501–507.

UNESCO (2021) Recommendation on the Ethics of Artificial Intelligence. Available at: [unesco_recommendation_on_the_ethics_of_artificial_intelligence_-_key_facts.pdf](#)